

Law of Cosines

New facts often trigger new ideas. – Alex F. Osborn

CHAPTER 18

Introduction to Trigonometry

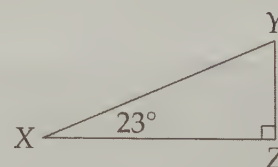
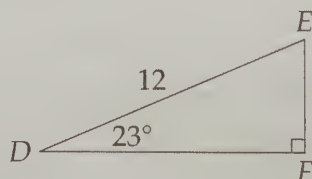
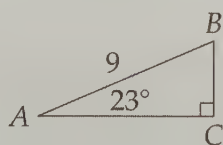
Trigonometry begins with the relationships among the angles and the sides of a triangle. We'll start by exploring these relationships for right triangles, and then we'll expand our study to all triangles.

The main purpose of this chapter is to give you a little taste of trigonometry, so that it won't be brand-new when you dive more deeply into the subject. We'll explore trigonometry in much more detail in later textbooks. Hopefully, at that time, this taste will make you a little more comfortable with trigonometry as you learn to apply it more broadly.

18.1 Trigonometry and Right Triangles

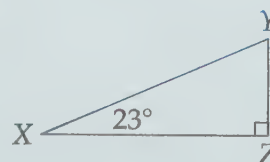
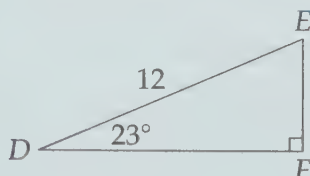
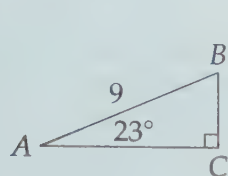
Problems

Problem 18.1: You may use a calculator on this problem.



- In the diagram above, $BC \approx 3.5$. What is EF to the nearest tenth?
- Use the given information about $\triangle ABC$ to find YZ/XY to the nearest 0.01.

Problem 18.1: You may use a calculator on this problem.



- (a) In the diagram above, $BC \approx 3.5$. What is EF to the nearest tenth?
- (b) Find YZ/XY to the nearest 0.01.

Solution for Problem 18.1:

- (a) We have $\triangle ABC \sim \triangle DEF$ by AA Similarity. Therefore, we have $BC/AB = EF/DE$, so

$$EF = (DE) \left(\frac{BC}{AB} \right) = (12) \left(\frac{3.5}{9} \right) \approx 4.7.$$

- (b) We have $\triangle ABC \sim \triangle XYZ$ by AA Similarity. Therefore, we have $YZ/XY = BC/AB \approx 0.39$. Notice that EF/DE is also approximately 0.39, which shouldn't be surprising because our triangle similarities mean that $YZ/XY = BC/AB = EF/DE$.

□

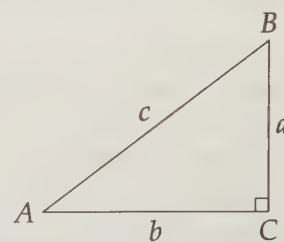
The ratio

$$\frac{\text{length of a leg opposite an acute angle of a right triangle}}{\text{length of the hypotenuse of the right triangle}}$$

is so common and important that we give it a name. This ratio is called the **sine** of the acute angle. We denote the sine of $\angle A$ as $\sin A$.

In $\triangle ABC$ at right, we have $\angle C = 90^\circ$, $BC = a$, $CA = b$, and $AB = c$. So, our definition of sine gives us

$$\sin A = \frac{\text{length of leg opposite } A}{\text{length of hypotenuse}} = \frac{a}{c}.$$



In Problem 18.1, we used similar triangles to show that in all right triangles with an acute angle of 23° , the ratio of the length of the leg opposite the 23° angle to the length of the hypotenuse is the same. Specifically, we found that this ratio is approximately 0.39. So, we write $\sin 23^\circ \approx 0.39$ to indicate that in any right triangle with an acute angle of 23° , the ratio

$$\frac{\text{length of the leg opposite the } 23^\circ \text{ angle}}{\text{length of the hypotenuse}}$$

is approximately 0.39.

We used similar triangles in Problem 18.1 to see that the ratio denoted by $\sin 23^\circ$ is the same for an acute angle measuring 23° in any right triangle. Similarly, we can show that for any value of x between 0° and 90° , the ratio denoted by $\sin x$ is the same in any right triangle with an acute angle measuring x .

This is why we define sine for an angle measure, not for a specific triangle – in any right triangle with acute angle 23° , the side length ratio referred to by $\sin 23^\circ$ is the same.

As noted earlier, we denote the sine of $\angle A$ as simply $\sin A$, rather than writing $\sin \angle A$. Because sine is a function of an angle measure, we'll assume that the $\angle$ symbol is implied when writing $\sin A$. However, we typically include the angle symbol when identifying an angle with three letters, so we'll write $\sin \angle XYZ$ rather than $\sin XYZ$. These conventions are common but not universal. Some sources will write $\sin XYZ$ instead of $\sin \angle XYZ$, and some write $\sin \angle A$ instead of $\sin A$.

The ratio denoted by sine is not the only useful ratio in a right triangle. The **cosine** of an acute angle in a right triangle equals

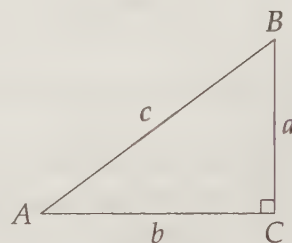
$$\frac{\text{length of the leg adjacent to the angle}}{\text{length of the hypotenuse}}.$$

We denote the cosine of $\angle A$ as $\cos A$, so in our diagram at right, we have

$$\cos A = \frac{\text{length of the leg adjacent to the angle}}{\text{length of the hypotenuse}} = \frac{b}{c}.$$

Finally, the **tangent** of an acute angle in a right triangle is the ratio of the sine of the angle to the cosine of the angle. We denote the tangent of $\angle A$ as $\tan A$, so we have

$$\tan A = \frac{\sin A}{\cos A}.$$



Sine, cosine, and tangent are examples of **trigonometric functions**. We often refer to problems involving trigonometric functions as an area of math called **trigonometry**.

Just as we did with $\sin x$, we can show that the ratio denoted by $\cos x$ is the same in any right triangle with an acute angle with measure x , as is the ratio denoted by $\tan x$.

Therefore, the value $\sin 23^\circ$ is a number, just like $\sqrt{7}$ is a number. The same is true for any trigonometric function of an acute angle. For most angles, we need to resort to a calculator to quickly find the value of the sine, cosine, or tangent of the angle. But there are some angles for which we can use our understanding of special right triangles to find the sine, cosine, and tangent of the angle without a calculator.

Problems

Problem 18.2: Show that the tangent of an acute angle in a right triangle equals the ratio

$$\frac{\text{length of the leg opposite the angle}}{\text{length of the leg adjacent to the angle}}.$$

Problem 18.3: Suppose $\triangle ABC$ is a 45-45-90 right triangle with $\angle C = 90^\circ$. Use $\triangle ABC$ to find $\sin 45^\circ$, $\cos 45^\circ$, and $\tan 45^\circ$.

Problem 18.4:

- (a) Find $\sin 30^\circ$, $\cos 30^\circ$, and $\tan 30^\circ$.
 (b) Find $\sin 60^\circ$, $\cos 60^\circ$, and $\tan 60^\circ$.

Problem 18.5: In $\triangle PQR$, we have $\angle P = 90^\circ$, $PQ = 3$, and $QR = 7$.

- (a) Find $\sin Q$ and $\cos R$.
 (b) Find $\tan Q$.
 (c) Find $(\sin R)^2 + (\cos R)^2$. Note: We often write $(\sin R)^2$ as $\sin^2 R$, and we do likewise for other positive powers of trigonometric functions.

Problem 18.6: Suppose that $\angle A$ is an acute angle.

- (a) Explain why $\sin A = \cos(90^\circ - A)$.
 (b) Explain why $\sin^2 A + \cos^2 A = 1$.

Problem 18.7: Let $\triangle ABC$ be a right triangle with $\angle C = 90^\circ$.

- (a) Show that $\sin A$ and $\cos A$ must be between 0 and 1.
 (b) Show that $\tan A$ must be greater than 0, but that $\tan A$ has no upper bound. In other words, show that for any positive number x , there is an angle A such that $\tan A = x$.

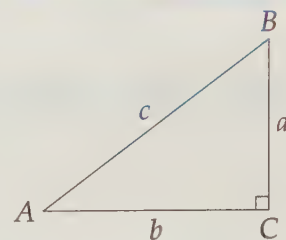
Problem 18.2: Show that the tangent of an acute angle in a right triangle equals the ratio

$$\frac{\text{length of the leg opposite the angle}}{\text{length of the leg adjacent to the angle}}.$$

Solution for Problem 18.2: Let $\triangle ABC$ be right triangle with $\angle C = 90^\circ$, $BC = a$, $AC = b$, and $AB = c$, as shown at right. Since tangent is the ratio of sine to cosine, we have

$$\tan A = \frac{\sin A}{\cos A} = \frac{\frac{a}{c}}{\frac{b}{c}} = \frac{a}{b} = \frac{\text{length of the leg opposite the angle}}{\text{length of the leg adjacent to the angle}}.$$

□



Putting the result of Problem 18.2 together with our definitions of sine and cosine, we have the following for right triangle $\triangle ABC$ above:

Important:

$$\begin{aligned}\sin A &= \frac{\text{length of leg opposite } A}{\text{length of hypotenuse}} = \frac{a}{c}, \\ \cos A &= \frac{\text{length of leg adjacent to } A}{\text{length of hypotenuse}} = \frac{b}{c}, \\ \tan A &= \frac{\text{length of leg opposite } A}{\text{length of leg adjacent to } A} = \frac{a}{b}.\end{aligned}$$

Since these are definitions, we have no choice but to simply memorize them. One common mnemonic for memorizing them is 'SOHCAHTOA':

Sine = Opposite/Hypotenuse; Cosine = Adjacent/Hypotenuse; Tangent = Opposite/Adjacent.

We can't quite do SOHCAHTOA justice in print – it's a much more effective memorization tool when spoken, "So-cah-toe-ah."

We can find the values of trigonometric functions of some angles with our understanding of special right triangles.

Problem 18.3: Find $\sin 45^\circ$, $\cos 45^\circ$, and $\tan 45^\circ$.

Solution for Problem 18.3: At right is 45-45-90 triangle ABC with right angle at C . The ratio of a leg of a 45-45-90 triangle to its hypotenuse is $1/\sqrt{2}$, so we have

$$\sin 45^\circ = \sin A = \frac{BC}{AB} = \frac{1}{\sqrt{2}}.$$

We usually try to avoid writing fractions with a square root in the denominator. We can write $1/\sqrt{2}$ without a square root in the denominator by multiplying the numerator and denominator by $\sqrt{2}$:

$$\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Similarly, we have $\cos 45^\circ = \cos A = \frac{AC}{AB} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. Finally, since the legs of a 45-45-90 triangle have the same length, we have

$$\tan 45^\circ = \tan A = \frac{BC}{AC} = 1.$$

□

Problem 18.4: Find $\sin 30^\circ$, $\cos 30^\circ$, and $\tan 30^\circ$. Find $\sin 60^\circ$, $\cos 60^\circ$, and $\tan 60^\circ$.

Solution for Problem 18.4: We start with $\triangle ABC$ at right, in which $\angle A = 30^\circ$, $\angle C = 90^\circ$, and $\angle B = 60^\circ$. Because $\triangle ABC$ is a 30-60-90 right triangle, we have $BC/AB = 1/2$, so

$$\sin 30^\circ = \frac{BC}{AB} = \frac{1}{2} \quad \text{and} \quad \cos 60^\circ = \frac{BC}{AB} = \frac{1}{2}.$$

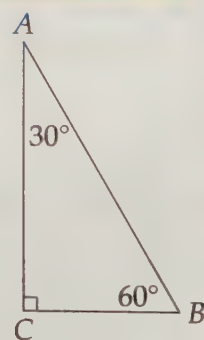
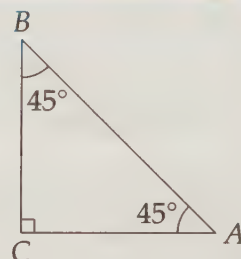
We also have $AC/AB = \sqrt{3}/2$, so

$$\sin 60^\circ = \frac{AC}{AB} = \frac{\sqrt{3}}{2} \quad \text{and} \quad \cos 30^\circ = \frac{AC}{AB} = \frac{\sqrt{3}}{2}.$$

Since $AC/BC = \sqrt{3}/1$, we have

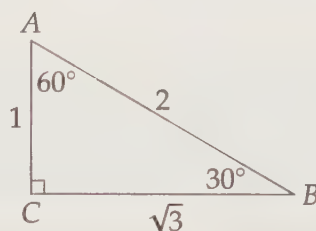
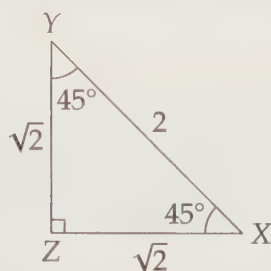
$$\tan 60^\circ = \frac{AC}{BC} = \sqrt{3} \quad \text{and} \quad \tan 30^\circ = \frac{BC}{AC} = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

□



The table below shows the values of sine, cosine, and tangent for the special angles we have explored in this section. The 45-45-90 triangle and the 30-60-90 triangle below exhibit the relationships among the sides of each triangle that we used to find these values.

Angle	sin	cos	tan
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$



You shouldn't have to memorize these! These are just another way of expressing the ratios of sides in 45-45-90 triangles and in 30-60-90 triangles.

If we have enough information about the sides of a not-so-special right triangle, we can evaluate the trigonometric functions of its angles, too.

Problem 18.5: In $\triangle PQR$, we have $\angle P = 90^\circ$, $PQ = 3$, and $QR = 7$.

- Find $\sin Q$ and $\cos R$.
- Find $\tan Q$.
- Find $(\sin R)^2 + (\cos R)^2$.

Solution for Problem 18.5:

- We first use the Pythagorean Theorem to determine that $PR = 2\sqrt{10}$. Then, we draw our diagram with the sides labeled as at right. We have $\sin Q = PR/QR = 2\sqrt{10}/7$ and $\cos R = PR/QR = 2\sqrt{10}/7$.
- We have

$$\tan Q = \frac{PR}{PQ} = \frac{2\sqrt{10}}{3}.$$

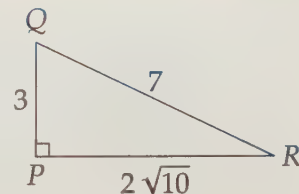
- We first find $\sin R = PQ/QR = 3/7$. Then, we have

$$(\sin R)^2 + (\cos R)^2 = \left(\frac{3}{7}\right)^2 + \left(\frac{2\sqrt{10}}{7}\right)^2 = \frac{9}{49} + \frac{40}{49} = 1.$$

□

We typically write $(\sin R)^2$ and $(\cos R)^2$ as $\sin^2 R$ and $\cos^2 R$, respectively, and similarly for other positive integer powers of trigonometric functions. So, we'd write the result we found in part (c) as $\sin^2 R + \cos^2 R = 1$.

Parts (a) and (c) above suggest a couple interesting relationships between trigonometric functions. Let's investigate.



Problem 18.6: Suppose that $\angle A$ is an acute angle.

- (a) Explain why $\sin A = \cos(90^\circ - A)$.
- (b) Explain why $\sin^2 A + \cos^2 A = 1$.

Solution for Problem 18.6:

- (a) At right is right triangle $\triangle ABC$ with acute angle $\angle A$. We have $\sin A = BC/AB$. We also have $\angle B = 90^\circ - \angle A$, so

$$\cos(90^\circ - A) = \cos B = \frac{BC}{AB}.$$

Therefore, we have $\sin A = \cos(90^\circ - A)$ for any acute angle $\angle A$.

- (b) We have $\sin A = BC/AB$ and $\cos A = AC/AB$, so

$$\sin^2 A + \cos^2 A = \frac{BC^2}{AB^2} + \frac{AC^2}{AB^2} = \frac{BC^2 + AC^2}{AB^2}.$$

The Pythagorean Theorem gives us $BC^2 + AC^2 = AB^2$, so we have

$$\sin^2 A + \cos^2 A = \frac{BC^2 + AC^2}{AB^2} = \frac{AB^2}{AB^2} = 1.$$

□

A **trigonometric identity** is a statement involving trigonometric functions that is true for all values of the angles in the statement. In Problem 18.6, we proved two trigonometric identities for acute angles $\angle A$:

Important:



$$\sin^2 A + \cos^2 A = 1,$$

$$\sin A = \cos(90^\circ - A).$$

Once again, note that $\sin^2 A = (\sin A)^2$ (and likewise for all trigonometric functions raised to positive integer powers).

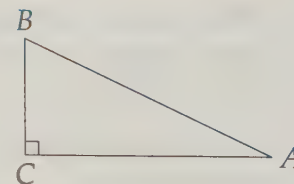
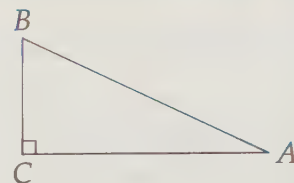
Problem 18.7: Let $\triangle ABC$ be a right triangle with $\angle C = 90^\circ$.

- (a) Show that $\sin A$ and $\cos A$ must be between 0 and 1.
- (b) Show that $\tan A$ must be greater than 0, but that $\tan A$ has no upper bound.

Solution for Problem 18.7:

- (a) In $\triangle ABC$ at right, we have $\sin A = BC/AB$. Because $\overline{AB}$ is the hypotenuse of $\triangle ABC$, it is longer than both legs. Specifically, we have $AB > BC$, so $BC/AB < 1$. Furthermore, both lengths are positive, so $BC/AB > 0$. Combining these inequalities gives us $0 < \sin A < 1$.

Similarly, we have $\cos A = AC/AB$ and $AB > AC > 0$, so $0 < \cos A < 1$.



- (b) Because $\tan A = BC/AC$ and BC and AC are positive, we have $\tan A > 0$. To see that $\tan A$ can take on any positive value, let $AC = 1$. Then, we have $\tan A = BC$. We have no restrictions on BC . For any positive number x , we can build a right triangle with legs $AC = 1$ and $BC = x$, so that $\tan A = x$. Therefore, $\tan A$ can take on any positive value, so there is no upper bound on the possible values of $\tan A$.

□

Important: If $\angle A$ is acute, then $0 < \sin A < 1$, $0 < \cos A < 1$, and $\tan A > 0$. There is no upper bound on the value of $\tan A$.



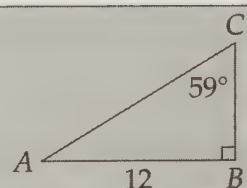
Note that while we proved that $\tan A$ can take on any positive value, we did not show that $\sin A$ and $\cos A$ can equal any positive number between 0 and 1. You'll have a chance to prove this in the Exercises.

We've seen that we can evaluate trigonometric functions for some special angles without a calculator. However, we use a calculator or a computer to find a value of the trigonometric functions of most angles. For the next set of problems, you are free to use your calculator.

WARNING!! Degrees are not the only units for angles. If you use your calculator to compute $\sin 30^\circ$ and you don't get 0.5 (or $1/2$) as the result, then either you aren't using your calculator correctly, or your calculator is set to use units besides degrees for angles. If your calculator is set to use some other units of measure besides degrees, reset your calculator to degrees before continuing.

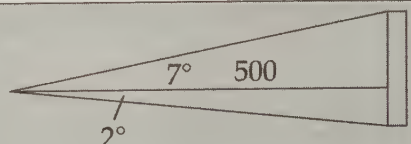
Problems

Problem 18.8: Find AC and BC in the diagram at right to the nearest tenth.

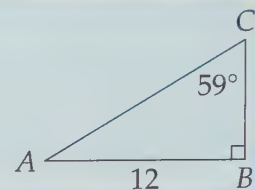


Problem 18.9: Right triangle ABC has $\angle ABC = 90^\circ$. Let the midpoint of $\overline{AB}$ be M . The perpendicular bisector of $\overline{AB}$ intersects $\overline{AC}$ at D , and $\sin \angle MDB = 1/5$. If $MD = 1$, then what is AC ?

Problem 18.10: I am 500 feet from a building. When I look at the top of the building, I look upward at an angle of 7° from horizontal. When I look at the point where the building touches the ground, I look downward at an angle of 2° from horizontal. How tall is the building to the nearest foot?



Problem 18.8: Find AC and BC in the diagram at right to the nearest tenth.



Solution for Problem 18.8: We have $\sin C = AB/AC$, so we have $AC = AB/(\sin C)$. Since $\sin C = \sin 59^\circ \approx 0.857$, we have

$$AC = \frac{AB}{\sin C} \approx \frac{12}{0.857} \approx 14.0.$$

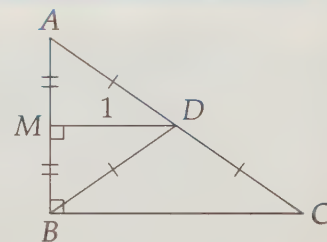
We can now use either the Pythagorean Theorem or trigonometry to find BC . Since we're in the trigonometry chapter, we'll use $\cos C$. Since $\cos C = BC/AC$, we have $BC = AC \cos C$. Since $C = 59^\circ$, we have $\cos 59^\circ \approx 0.515$, so $BC \approx (14.0)(.515) \approx 7.2$. $\square$

Problem 18.9: Right triangle ABC has $\angle ABC = 90^\circ$. Let the midpoint of $\overline{AB}$ be M . The perpendicular bisector of $\overline{AB}$ intersects $\overline{AC}$ at D , and $\sin \angle MDB = 1/5$. If $MD = 1$, then what is AC ?

Solution for Problem 18.9: Oh no! There's no diagram! What will we do?

We start by drawing our own diagram. Don't let wordy, seemingly complicated geometry problems scare you. Take the time to draw a diagram for the problem, label it with information you know, then get to work. As you discover more about the diagram, add what you learn to the diagram.

Here, we start with the perpendicular bisector of $\overline{AB}$. The perpendicular bisectors of a triangle pass through the circumcenter of the triangle. The circumcenter of a right triangle is the midpoint of its hypotenuse. So, point D is the midpoint of $\overline{AC}$. Therefore, if we find AD , DC , or BD , we can easily find AC .



We focus on the information we haven't used yet: $MD = 1$ and $\sin \angle MDB = 1/5$. We'd like to find BD . From $\sin \angle MDB = 1/5$, we have

$$\frac{MB}{BD} = \frac{1}{5}.$$

Unfortunately, we don't know MB . However, we do know DM , and we know that

$$\frac{DM}{BD} = \cos \angle MDB.$$

So, if we can find $\cos \angle MDB$, we can find BD from this equation. We aren't given $\cos \angle MDB$, but we do know $\sin \angle MDB$, and we know that

$$\cos^2 \angle MDB + \sin^2 \angle MDB = 1.$$

Therefore, we have $\cos^2 \angle MDB = 1 - \sin^2 \angle MDB = 24/25$. We must have $\cos \angle MDB > 0$, so we have $\cos \angle MDB = 2\sqrt{6}/5$. From $DM/BD = \cos \angle MDB$, we have $BD/DM = 1/(\cos \angle MDB)$, so

$$BD = \frac{DM}{\cos \angle MDB} = \frac{5}{2\sqrt{6}} = \frac{5}{2\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{5\sqrt{6}}{12}.$$

Finally, we have

$$AC = 2AD = 2BD = \frac{5\sqrt{6}}{6}.$$

□

Our process of finding BD using $MD = 1$ and $\sin \angle MDB = 1/5$ in right triangle MDB is an example of the fact that:

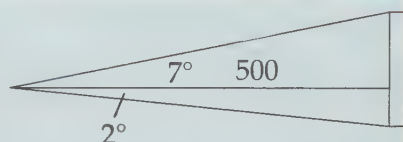
Important:



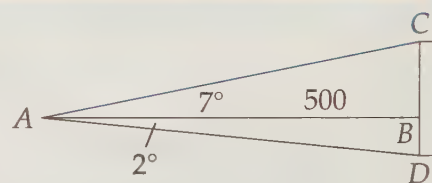
If we have the length of a side of a right triangle and the value of a trigonometric function of either of the acute angles of the triangle, then we can find all the side lengths of the triangle.

Trigonometric functions are used for surveying land and measuring large distances. Here's an example how.

Problem 18.10: I am 500 feet from a building. When I look at the top of the building, I look upward at an angle of 7° from horizontal. When I look at the point where the building touches the ground, I look downward at an angle of 2° from horizontal. How tall is the building to the nearest foot?



Solution for Problem 18.10: We label the diagram as shown at right. We seek the length of $\overline{CD}$. While we can't find CD directly, we can use trigonometry to find BC and BD . Specifically, we have $\tan \angle CAB = BC/BA$, so $BC = BA \tan 7^\circ \approx 61.4$. We also have $\tan \angle BAD = BD/BA$, so $BD = BA \tan 2^\circ \approx 17.5$. Therefore, to the nearest foot, the height of the building is $CD = CB + BD \approx 79$ feet. □



Sine, cosine, and tangent are not the only trigonometric functions. Three more trigonometric functions are **cosecant** (denoted $\csc$), **secant** (denoted $\sec$), and **cotangent** (denoted $\cot$). These are defined as follows:

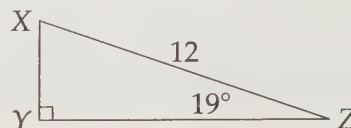
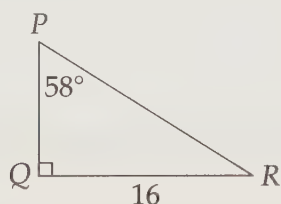
$$\begin{aligned}\csc x &= \frac{1}{\sin x} \\ \sec x &= \frac{1}{\cos x} \\ \cot x &= \frac{1}{\tan x}\end{aligned}$$

There's no need to memorize these right now – you'll have enough experience with them in later texts that you'll come to know them well.

Exercises

18.1.1 Let $\triangle ABC$ be a right triangle with $\angle ABC = 90^\circ$, $AB = 12$, and $AC = 16$. Find $\sin A$, $\cos A$, and $\tan A$.

18.1.2 Find the missing sides in the triangles below to the nearest 0.1.



18.1.3 Let x be a real number such that $0 < x < 1$. Prove that there exists an acute $\angle A$ such that $\sin A = x$, and that there exists an acute angle $\angle B$ such that $\cos B = x$. **Hints:** 1

18.1.4 Show that if θ is the measure of an acute angle, then $\cos \theta = \sin(90^\circ - \theta)$.

18.1.5 In triangle ABC , we have $\angle B = 90^\circ$ and $\sin A = 5/7$. Find $\tan C$.

18.1.6 Let $\overline{AB}$ be a diameter of $\odot O$, where $AB = 2$. Suppose $\overline{AX}$ is tangent to $\odot O$, and that $\overline{AY}$ bisects $\angle XAO$. The angle bisector of $\angle YAB$ intersects $\odot O$ at point Z (where A and Z are different points). Find AZ to the nearest hundredth.

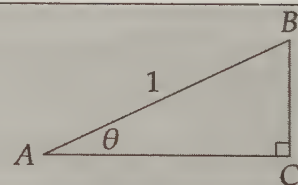
18.1.7★ Without using a calculator, evaluate $\sin 15^\circ$. **Hints:** 513, 536

18.2 Not Just For Right Triangles

So far, we have only defined trigonometric functions for angles in right triangles. However, these functions are useful for far more than just right triangles. But to see why, we'll start with a special right triangle.

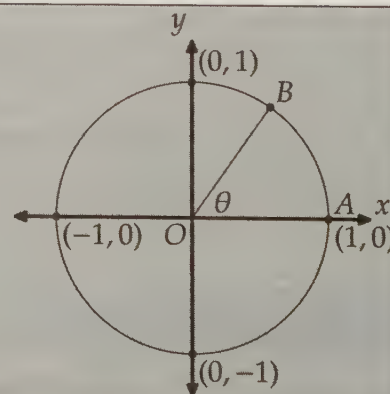
Problems

Problem 18.11: In $\triangle ABC$ at right, we have $\angle C = 90^\circ$, $AB = 1$, and $\angle A = \theta$. Find expressions for AC and BC in terms of θ .



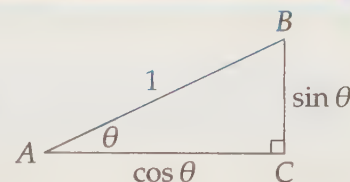
Problem 18.12: We commonly refer to the circle centered at the origin with radius 1 as the **unit circle**, which is shown at right.

Let O be the origin, A be the point $(1, 0)$ and B be a point on the unit circle such that both coordinates of B are positive and $\angle BOA = \theta$. Find the coordinates of B in terms of θ .



Problem 18.11: In $\triangle ABC$, let $\angle C = 90^\circ$, $AB = 1$, and $\angle A = \theta$. Find expressions for AC and BC in terms of θ .

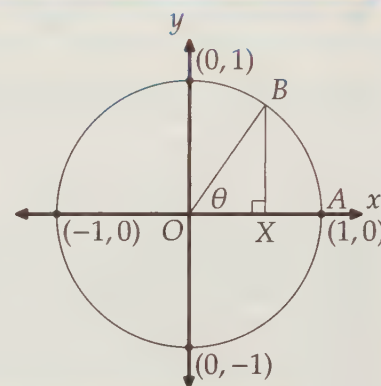
Solution for Problem 18.11: Because $AB = 1$, we have $\sin A = BC/AB = BC$. Therefore, $BC = \sin A = \sin \theta$. Similarly, we have $\cos A = AC/AB = AC$, so $AC = \cos A = \cos \theta$. These side lengths are shown in the diagram at right. $\square$



We can extend our observation in Problem 18.11 to build a definition of sine and cosine for angles that are not acute. First, we'll take a slightly different look at depicting the cosine and sine of an acute angle.

Problem 18.12: We commonly refer to the circle centered at the origin with radius 1 as the **unit circle**. Let O be the origin, A be the point $(1, 0)$ and B be a point on the unit circle such that both coordinates of B are positive and $\angle BOA = \theta$. Find the coordinates of B in terms of θ .

Solution for Problem 18.12: To find the coordinates of B in terms of θ , we draw altitude $\overline{BX}$ from B to the x -axis. Doing so builds a right triangle with an acute angle with measure θ . Furthermore, the hypotenuse, $\overline{OB}$, of this triangle is a radius of the unit circle, so $OB = 1$. Now, we have the same problem as in Problem 18.11, and we have $OX = \cos \theta$ and $XB = \sin \theta$. Therefore, the coordinates of B are $(\cos \theta, \sin \theta)$. $\square$

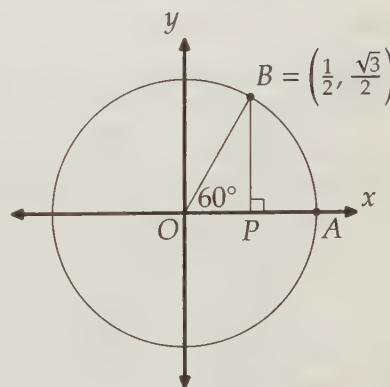


Rather than using right triangles to define cosine and sine, we can extend our observation in Problem 18.12 to use the unit circle to define cosine and sine.

Important: Let point A be $(1, 0)$ and point B be on the unit circle such that B is θ degrees counterclockwise from A . Then, we define $\cos \theta$ and $\sin \theta$ to be the x -coordinate and y -coordinate, respectively, of point B .

For example, point B at right is on the unit circle, 60° counterclockwise from $(1, 0)$. Drawing altitude $\overline{BP}$ from B to the x -axis forms 30-60-90 triangle $\triangle BPO$, from which we have $BO = 1/2$ and $BP = \sqrt{3}/2$. Therefore, the coordinates of B are $(1/2, \sqrt{3}/2)$, so we have $\cos 60^\circ = 1/2$ and $\sin 60^\circ = \sqrt{3}/2$.

Let's see how this definition allows us to define sine and cosine of angles that are not acute.



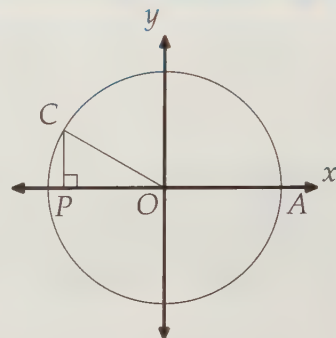
Problems

Problem 18.13: Let point A be $(1, 0)$ and let point C be on the unit circle such that $\widehat{AC} = 150^\circ$ and such that C is 150° counterclockwise from A along the circle. Find the coordinates of C .

Problem 18.14: Let point A be $(1, 0)$ and let point D be on the unit circle such that $\widehat{AD} = 225^\circ$ and such that D is 225° counterclockwise from A along the circle. Find the coordinates of D .

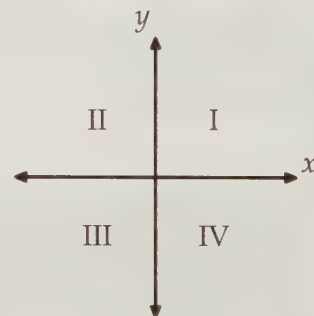
Problem 18.13: Find $\cos 150^\circ$ and $\sin 150^\circ$.

Solution for Problem 18.13: Following our definition of cosine and sine, we let A be $(1, 0)$ and we let point C be the point on the unit circle that is 150° counterclockwise from A . The coordinates of C then are $(\cos 150^\circ, \sin 150^\circ)$. To find these coordinates, we first determine where point C is along the unit circle. The point $(0, 1)$ is 90° counterclockwise from A and $(-1, 0)$ is 180° counterclockwise from A , so C is on the minor arc from $(0, 1)$ to $(-1, 0)$. At right, we have our unit circle, the origin, O , and points A and C .



Once again, we build a right triangle by drawing the altitude from C to the x -axis. Because $\widehat{AC} = 150^\circ$, we have $\angle AOC = 150^\circ$. Therefore, we have $\angle COP = 180^\circ - 150^\circ = 30^\circ$. So, $\triangle COP$ is a 30-60-90 triangle. Since $OC = 1$ and $\angle COP = 30^\circ$, we have $OP = \sqrt{3}/2$ and $CP = 1/2$. Because C is to the left of the y -axis and above the x -axis, its x -coordinate is negative and its y -coordinate is positive. So, the coordinates of C are $(-\sqrt{3}/2, 1/2)$, which means $\cos 150^\circ = -\sqrt{3}/2$ and $\sin 150^\circ = 1/2$. $\square$

The coordinate axes divide the Cartesian plane into four **quadrants**. We usually label these quadrants with Roman numerals, as shown at right. We refer to them as the first, second, third, and fourth quadrant, respectively. So, point C in Problem 18.13 is in the second quadrant. All points in the second quadrant are to the left of the x -axis and above the y -axis, so all points in the second quadrant have a negative x -coordinate and a positive y -coordinate.

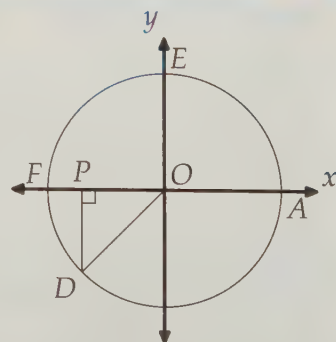


We sometimes refer to angle measures with quadrants, as well. When we say that a 150° angle is a 'second quadrant' angle, this means that the point that is 150° counterclockwise from $(1, 0)$ on the unit circle is in the second quadrant. Similarly, because 60° is between 0° and 90° , a 60° angle is a 'first quadrant' angle.

Problem 18.14: Evaluate $\cos 225^\circ$ and $\sin 225^\circ$.

Solution for Problem 18.14: Again, we let A be $(1, 0)$, and we let point D be the point on the unit circle that is 225° counterclockwise from A . Since $(-1, 0)$ is 180° counterclockwise from A and $(0, -1)$ is 270° counterclockwise from A , we know that D is in quadrant III, as shown at right.

We draw altitude $\overline{DP}$ from D to the x -axis, forming right triangle DOP . Because $\widehat{AEF}$ is 180° and $\widehat{AED}$ is 225° , we have $\widehat{DF} = 45^\circ$, so $\triangle DOP$ is a 45-45-90 triangle. Since $OD = 1$, we have $OP = DP = \sqrt{2}/2$. Point D is $\sqrt{2}/2$ to the left of the y -axis and $\sqrt{2}/2$ below the x -axis, so its coordinates are $(-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore, we have $\cos 225^\circ = \sin 225^\circ = -\sqrt{2}/2$. $\square$



Now that we have a handle on sine and cosine, we can revisit tangent. Once we have defined sine

and cosine in terms of the unit circle, we define the tangent of any angle θ for which $\cos \theta \neq 0$ as

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

If $\cos \theta = 0$, then $\tan \theta$ is undefined.

Problems

Problem 18.15:

- Use the unit circle to find $\sin 0^\circ$ and $\cos 0^\circ$, then evaluate $\tan 0^\circ$.
- Find $\sin 90^\circ$, $\cos 90^\circ$, and $\tan 90^\circ$.

Problem 18.16: Evaluate $\cos 300^\circ$, $\sin 300^\circ$, and $\tan 300^\circ$ without a calculator.

Problem 18.17: Explain why $\sin(180^\circ - \theta) = \sin \theta$ for any acute angle θ .

Problem 18.18: Show that for any triangle $\triangle ABC$, we have $[ABC] = \frac{1}{2}(AC)(BC) \sin C$.

Problem 18.15:

- Find $\sin 0^\circ$, $\cos 0^\circ$, then evaluate $\tan 0^\circ$.
- Find $\sin 90^\circ$, $\cos 90^\circ$, and $\tan 90^\circ$.

Solution for Problem 18.15:

- The point that is 0° counterclockwise from $(1, 0)$ is simply $(1, 0)$. Therefore, we have $\cos 0^\circ = 1$ and $\sin 0^\circ = 0$, so $\tan 0^\circ = (\sin 0^\circ)/(\cos 0^\circ) = 0$.
- The point that is 90° counterclockwise from $(1, 0)$ is $(0, 1)$. So, we have $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$. Therefore, we have $\tan 90^\circ = (\sin 90^\circ)/(\cos 90^\circ) = 1/0 = \dots$ Uh-oh! We can't divide by 0, so $\tan 90^\circ$ is undefined.

□

Sidenote: When we put the values of sine and cosine of 0° and 90° together with those of 30° , 45° , and 60° , a curious pattern emerges:

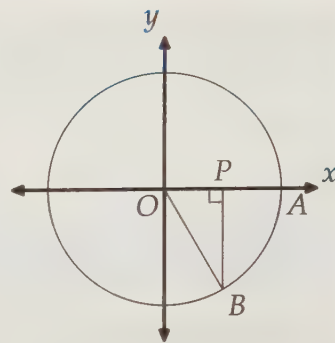


	0°	30°	45°	60°	90°
sin	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
cos	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$

Problem 18.16: Evaluate $\cos 300^\circ$, $\sin 300^\circ$, and $\tan 300^\circ$ without a calculator.

Solution for Problem 18.16: We let A be $(1, 0)$, and we let point B be the point on the unit circle that is 300° counterclockwise from A . The point $(0, -1)$ is 270° counterclockwise from A , and going 360° counterclockwise from A takes us back to A , so we know that B is in quadrant IV, as shown at right. (We could also have determined that B is in quadrant IV by noting that 300° counterclockwise is the same as 60° clockwise.)

We draw the altitude from B to point P on the x -axis. Because B is 300° counterclockwise from A , we have $\widehat{AB} = 60^\circ$, so $\angle BOA = 60^\circ$. From 30-60-90 triangle BPO , we have $BP = \sqrt{3}/2$ and $OP = 1/2$, so the coordinates of B are $(1/2, -\sqrt{3}/2)$. What's wrong with this conclusion:



Bogus Solution: Because B is $(1/2, -\sqrt{3}/2)$, we have $\sin 300^\circ = 1/2$ and $\cos 300^\circ = -\sqrt{3}/2$.



We have sine and cosine backwards here! Cosine is the x -coordinate and sine is the y -coordinate.

WARNING!! Be careful not to get sine and cosine backwards. Once you have found the appropriate point on the unit circle, an easy way to remember which coordinate is cosine and which is sine is to use alphabetical order: cosine is before sine and x is before y .

So, because B is $(1/2, -\sqrt{3}/2)$, we have $\cos 300^\circ = 1/2$ and $\sin 300^\circ = -\sqrt{3}/2$. Therefore, we have

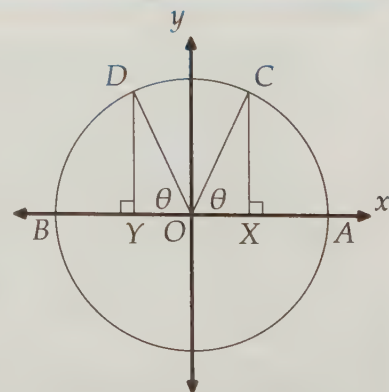
$$\tan 300^\circ = \frac{\sin 300^\circ}{\cos 300^\circ} = \frac{-\sqrt{3}/2}{1/2} = -\sqrt{3}.$$

□

We can use the unit circle to 'see' trigonometric relationships.

Problem 18.17: Explain why $\sin(180^\circ - \theta) = \sin \theta$ for any acute angle θ .

Solution for Problem 18.17: Let A be $(1, 0)$ and B be $(-1, 0)$. Let C be $(\cos \theta, \sin \theta)$, so that C is θ counterclockwise from A . Let D be the point $(\cos(180^\circ - \theta), \sin(180^\circ - \theta))$, so D is $180^\circ - \theta$ counterclockwise from A . Because $(-1, 0)$ is 180° counterclockwise from A and D is $180^\circ - \theta$ counterclockwise from A , point D is θ clockwise from $(-1, 0)$, as shown in the diagram. We draw altitudes $\overline{CX}$ and $\overline{DY}$ to the x -axis, since these lengths equal $\sin \theta$ and $\sin(180^\circ - \theta)$, respectively. Because $OD = OC = 1$, we have $\triangle OYD \cong \triangle OXC$ by SA Congruence for right triangles. This gives us $CX = DY$, so $\sin \theta = \sin(180^\circ - \theta)$. □



With a little bit of casework, we can extend this proof to show that $\sin(180^\circ - \theta) = \sin \theta$ for all angles θ , not just acute angles. This relationship is not worth memorizing. After working with trigonometric functions and the unit circle more, this

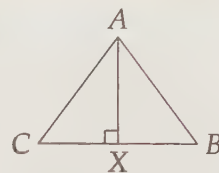
relationship, and others like it, will become automatic to you. We'll be exploring how in later books in the *Art of Problem Solving* series.

Problem 18.18: Show that for any triangle $\triangle ABC$, we have $[ABC] = \frac{1}{2}(AC)(BC) \sin C$.

Solution for Problem 18.18: We consider separately the cases in which C is acute, right, or obtuse.

Case 1: $\angle C$ is acute. To find the area, we seek an expression for the altitude from A to $\overline{BC}$. So, we draw altitude $\overline{AX}$, forming right triangle $\triangle AXC$. From this triangle, we have $\sin C = AX/AC$, so $AX = AC \sin C$. Therefore, we have

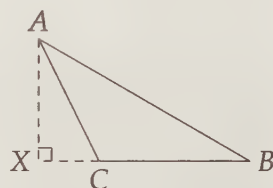
$$[ABC] = \frac{1}{2}(AX)(BC) = \frac{1}{2}(AC)(BC) \sin C.$$



Case 2: $\angle C$ is right. If $\angle C$ is right, then we have both $\sin C = 1$ and $[ABC] = \frac{1}{2}(AC)(BC)$. So, we have $[ABC] = \frac{1}{2}(AC)(BC) \sin C$.

Case 3: $\angle C$ is obtuse. If $\angle C$ is obtuse, then altitude $\overline{AX}$ is outside $\triangle ABC$, as shown at right. We have $\sin \angle ACX = AX/AC$, so $AX = AC \sin \angle ACX$. Because $\angle ACX = 180^\circ - \angle ACB$, we have

$$[ABC] = \frac{1}{2}(AX)(BC) = \frac{1}{2}(AC)(BC) \sin \angle ACX = \frac{1}{2}(AC)(BC) \sin(180^\circ - \angle ACB).$$



Because $\sin(180^\circ - \angle ACB) = \sin \angle ACB$ for any angle $\angle ACB$, we have $[ABC] = \frac{1}{2}(AC)(BC) \sin \angle ACB$, as desired.

We have covered all three possible cases for $\angle C$, so we have shown that

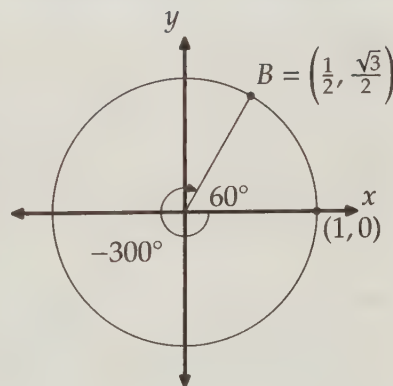
Important: Let $BC = a$ and $AC = b$ in $\triangle ABC$. Then, we have



$$[ABC] = \frac{1}{2}ab \sin C.$$

□

Using our unit circle, we can also evaluate trigonometric functions for angle measures greater than 360° , as well as for negative angle measures. For example, since going 360° counterclockwise from $(1, 0)$ brings us back to $(1, 0)$, the point that is $360^\circ + 60^\circ = 420^\circ$ counterclockwise from $(1, 0)$ at right is point B shown. From the coordinates of B , we see that $\cos 420^\circ = 1/2$ and $\sin 420^\circ = \sqrt{3}/2$. To evaluate trigonometric functions of negative angles, we go clockwise from $(1, 0)$ rather than counterclockwise. So, because B is 300° clockwise from $(1, 0)$, we have $\cos(-300^\circ) = 1/2$ and $\sin(-300^\circ) = \sqrt{3}/2$.



Exercises

18.2.1 Evaluate each of the following:

- | | | |
|----------------------|-----------------------|----------------------|
| (a) $\sin 120^\circ$ | (c) $\tan 135^\circ$ | (e) $\cos 210^\circ$ |
| (b) $\sin 330^\circ$ | (d) $\cos(-45^\circ)$ | (f) $\tan 720^\circ$ |

18.2.2 Explain why $\sin^2 \theta + \cos^2 \theta = 1$ for any angle measure θ .

18.2.3 How many angle measures θ such that $0 \leq \theta < 360^\circ$ satisfy the equation $\sin \theta = 0.31$?

18.2.4 Explain why $\cos(360^\circ + \theta) = \cos \theta$ for any angle θ .

18.2.5 Explain why $\cos(180^\circ - \theta) = -\cos \theta$ for any acute angle θ .

18.3 Law of Sines and Law of Cosines

SAS Congruence tells us that if two sides of one triangle and the angle between them are equal to the corresponding sides and angle of another triangle, then the two triangles are congruent. Therefore, the third sides of these two triangles have the same length. In other words, if we know two side lengths of a triangle and the measure of the angle between those sides, then there is only one possible length of the third side of the triangle. Hmmm... Can we find that length if we know the two side lengths and the angle measure?

Similarly, ASA and AAS Congruence tell us that if two angles and a side of a triangle equal the corresponding angles and side of another triangle, then the two triangles are congruent. In other words, if we know two angles and a side in one triangle, then there is only one possible value for each of the remaining two side lengths. Can we find these two side lengths if we know the angles of the triangle and the other side length?

In this section, we use trigonometry to answer both of these questions.

Note: you can use your calculator for the problems in this section.

Problems

Problem 18.19: In $\triangle ABC$, let $AC = 15$, $BC = 12$, and $\angle C = 34^\circ$. In this problem, we find AB to the nearest hundredth.

- Draw altitude $\overline{BX}$ from B to $\overline{AC}$. Find CX and BX to the nearest hundredth.
- Find XA to the nearest hundredth.
- Use parts (a) and (b) to find AB to the nearest hundredth.

Extra! *Mathematical knowledge adds vigor to the mind, frees it from prejudice, credulity, and superstition.*

—John Arbuthnot

Problem 18.20: Let $\triangle ABC$ be an acute triangle with $a = BC$, $b = AC$, and $c = AB$. In this problem, we prove that

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

- Draw altitude $\overline{BX}$ from B to $\overline{AC}$. Express BX and CX in terms of a , b , and/or trigonometric functions of $\angle C$.
- Express AX in terms of a , b , and/or trigonometric functions of $\angle C$.
- Use $\triangle ABX$ to show that $c^2 = a^2 + b^2 - 2ab \cos C$.
- What happens in the equation $c^2 = a^2 + b^2 - 2ab \cos C$ if $\angle C$ is a right angle?

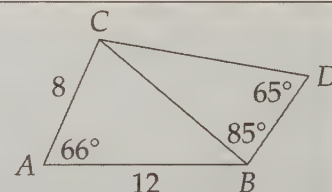
Problem 18.21: In $\triangle PQR$, suppose $PR = 12$, $\angle QPR = 66^\circ$, and $\angle PRQ = 63^\circ$. In this problem, we find PQ and QR .

- Let T be the foot of the altitude from P to $\overline{QR}$. Find PT to the nearest hundredth.
- Use PT to find PQ to the nearest hundredth.
- Find QR to the nearest hundredth.

Problem 18.22: Suppose that $\triangle ABC$ is an acute triangle with $a = BC$, $b = AC$, and $c = AB$. Prove that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Problem 18.23: Find BC , CD , and BD to the nearest tenth in the diagram at right.



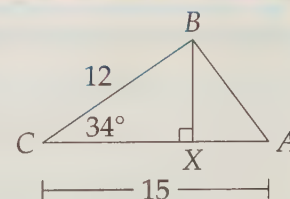
Problem 18.19: In $\triangle ABC$, let $AC = 15$, $BC = 12$, and $\angle C = 34^\circ$. Find AB to the nearest hundredth.

Solution for Problem 18.19: It's not immediately obvious how to find AB . We know how to use trigonometry to find side lengths in right triangles, so we start by drawing an altitude from B to $\overline{AC}$ as shown. This creates right triangle $\triangle BCX$ with the 34° angle as one of its acute angles. From right triangle $\triangle CBX$, we have

$$\frac{BX}{BC} = \sin C \approx 0.559,$$

so $BX \approx 0.559(BC) \approx 6.71$. Similarly, we have $CX/BC = \cos C \approx 0.829$, so $CX \approx 9.95$. This doesn't tell us AB yet, but we now have the length of one leg of $\triangle BXA$. If we can find the other, we can use the Pythagorean Theorem to find AB . Fortunately, XA is easy to find: $XA = AC - CX \approx 5.05$. Now, we can use the Pythagorean Theorem to find

$$AB = \sqrt{BX^2 + XA^2} \approx 8.40.$$

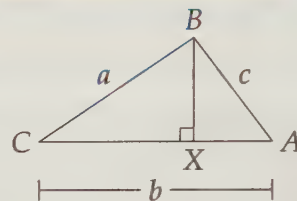


□

In Problem 18.19, we were given two side lengths of a triangle, and the measure of the angle between these two sides. We then found the third side. There was nothing particularly special about the side lengths or the angle. We might be able to follow essentially the same process for any triangle. Let's give it a try.

Problem 18.20: Let $\triangle ABC$ be an acute triangle with $a = BC$, $b = AC$, and $c = AB$. Suppose we know a , b , and $\angle C$. Find a formula that we can use to find c .

Solution for Problem 18.20: We use Problem 18.19 as a guide. In fact, this problem is essentially the same as Problem 18.19, but with variables a , b , and $\angle C$ in place of the numbers that were given in that problem. We can use the same diagram, and we'll use the same steps. We draw altitude $\overline{BX}$ from B to $\overline{AC}$. Then, we have $\sin C = BX/BC$, so $BX = BC \sin C = a \sin C$. We also have $\cos C = CX/BC$, so $CX = BC \cos C = a \cos C$. Therefore, we have $AX = AC - CX = b - a \cos C$. Next we apply the Pythagorean Theorem to $\triangle ABX$ to find $AB^2 = BX^2 + AX^2$. Substituting our expressions for these three sides gives us



$$\begin{aligned} c^2 &= a^2 \sin^2 C + (b - a \cos C)^2 \\ &= a^2 \sin^2 C + b^2 - 2ab \cos C + a^2 \cos^2 C \\ &= a^2(\sin^2 C + \cos^2 C) + b^2 - 2ab \cos C. \end{aligned}$$

Since $\sin^2 C + \cos^2 C = 1$, we have

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

□

We can test our new formula with the data from Problem 18.19.

Concept: Check formulas you prove by trying them on specific examples you have solved without the formula.



In Problem 18.19, we have $a = 12$, $b = 15$, and $\angle C = 34^\circ$, so we have

$$c^2 = a^2 + b^2 - 2ab \cos C \approx 70.546.$$

Taking the square root of both sides gives $c \approx 8.40$, which agrees with our answer from Problem 18.19.

With a little more casework (which you'll supply as an Exercise), we can show that this equation holds for any triangle ABC , not just for acute triangles.

Important: Let $a = BC$, $b = AC$, and $c = AB$ in $\triangle ABC$. The **Law of Cosines** states that



$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Notice that when $\angle C = 90^\circ$ in the Law of Cosines, we have $\cos C = 0$, so the law becomes $c^2 = a^2 + b^2$, which is just the Pythagorean Theorem.

Problem 18.19 is just a specific example of Problem 18.20, so we call Problem 18.20 a **generalization** of Problem 18.19.

Concept: Whenever we have to prove a general statement like the Law of Cosines, it's often useful to first consider a specific example of the general statement. If we can solve a specific example, we can sometimes use that solution as a guide to prove the generalization.

Now that you've learned the Law of Cosines, you're probably thinking there must be a Law of Sines, too. Of course, you've read the section title, so you know you're right. We'll discover the Law of Sines the same way we discovered the Law of Cosines – we'll start with a specific example.

Problem 18.21: Suppose $PR = 12$, $\angle QPR = 66^\circ$, and $\angle PRQ = 63^\circ$ in $\triangle PQR$. Find PQ and RQ to the nearest hundredth.

Solution for Problem 18.21: We start by building a right triangle, since we know how to use trigonometry to find lengths of sides in a right triangle. We draw altitude $\overline{PT}$ to side $\overline{QR}$. From right triangle $\triangle PTR$, we have $PT/PR = \sin R$, so $PT = PR \sin R \approx 10.69$. We also have $TR/PR = \cos R$, so $TR = PR \cos R \approx 5.45$.

Now, we can use right triangle $\triangle PQT$ to find lengths QT and PQ . First, we find that $\angle Q = 180^\circ - 66^\circ - 63^\circ = 51^\circ$. We have $PT/PQ = \sin Q$, so we have $PQ = PT/(\sin Q) \approx 13.76$. We also have $PT/QT = \tan Q$, so $QT = PT/(\tan Q) \approx 8.66$. Finally, we have $QR = QT + TR \approx 14.11$. $\square$

Let's try using our approach in Problem 18.21 to produce another 'law.'

Problem 18.22: Suppose that $\triangle ABC$ is an acute triangle with $a = BC$, $b = AC$, and $c = AB$. Prove that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Solution for Problem 18.22: We use Problem 18.21 as a guide. We draw altitude $\overline{AT}$ to $\overline{BC}$. From right triangle $\triangle ATC$, we have $\sin C = AT/AC$, so $AT = AC \sin C = b \sin C$. From right triangle $\triangle ABT$, we have $\sin B = AT/AB$, so $AT = AB \sin B = c \sin B$. These two expressions for AT must be equal, so we have $b \sin C = c \sin B$. Dividing this equation by $\sin B$ and by $\sin C$, we have

$$\frac{b}{\sin B} = \frac{c}{\sin C}.$$

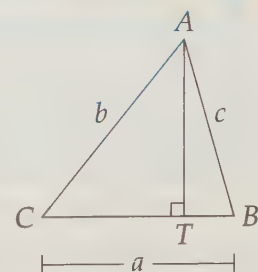
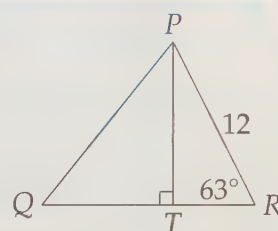
We can follow essentially the same steps starting with the altitude from C to $\overline{AB}$ to show that $b/(\sin B) = a/(\sin A)$. So, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$\square$

Important: Let $a = BC$, $b = AC$, and $c = AB$ in $\triangle ABC$. The **Law of Sines** states that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$



As with the Law of Cosines, you'll tackle the cases in which $\triangle ABC$ is right or obtuse in the Exercises.

Sidenote: The Law of Sines gives us another look at why 'SSA Congruence' is not a valid theorem to prove triangle congruence. Suppose we have $\angle A = 30^\circ$, $BC = 3$, and $AC = 4$ in $\triangle ABC$. The Law of Sines then gives us

$$\frac{BC}{\sin A} = \frac{AC}{\sin B},$$

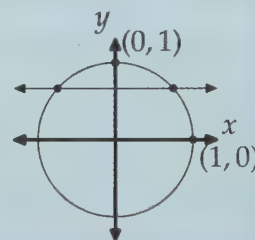
from which we find $\sin B = (AC/BC) \sin A = 2/3$.

Next, suppose we have $\angle D = 30^\circ$, $EF = 3$, and $DF = 4$ in $\triangle DEF$. Again, the Law of Sines gives us

$$\frac{EF}{\sin D} = \frac{DF}{\sin E},$$

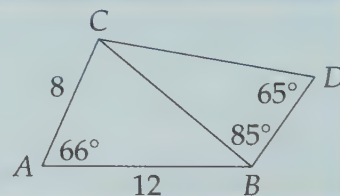
from which we find $\sin E = (DF/EF) \sin D = 2/3$. So, we have $\sin B = \sin E = 2/3$. Do we have $\angle B = \angle E$?

Not necessarily! There are two angles θ for which $0^\circ < \theta < 180^\circ$ and $\sin \theta = 2/3$. The two intersections of the graph of $y = 2/3$ with the unit circle correspond to these two angles; these are shown at right. So, we cannot deduce that $\angle B = \angle E$, and therefore we cannot conclude that $\triangle ABC \cong \triangle DEF$.



Let's give the Law of Cosines and the Law of Sines a try:

Problem 18.23: Find BC , CD , and BD to the nearest tenth in the diagram at right.



Solution for Problem 18.23: In $\triangle ABC$, we know two side lengths and the angle between these sides, so we can use the Law of Cosines to find BC . This gives us

$$BC^2 = 8^2 + 12^2 - 2(8)(12) \cos 66^\circ \approx 129.9,$$

so $BC \approx 11.4$. Turning to $\triangle BCD$, we have $\angle BCD = 180^\circ - 65^\circ - 85^\circ = 30^\circ$. Now, we know one of the side lengths of $\triangle BCD$ and all of the angle measures of $\triangle BCD$, so we can use the Law of Sines to find the missing side lengths. Specifically, we have

$$\frac{BC}{\sin D} = \frac{BD}{\sin \angle BCD} = \frac{CD}{\sin \angle CBD},$$

which means we have

$$\frac{11.4}{\sin 65^\circ} = \frac{BD}{\sin 30^\circ} = \frac{CD}{\sin 85^\circ}.$$

Solving for BD , we have

$$BD = \frac{11.4}{\sin 65^\circ} \cdot \sin 30^\circ \approx 6.3,$$

and solving for CD gives

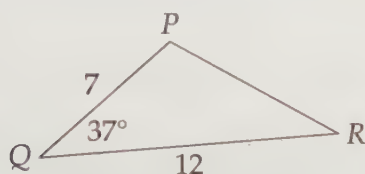
$$CD = \frac{11.4}{\sin 65^\circ} \cdot \sin 85^\circ \approx 12.5.$$

□

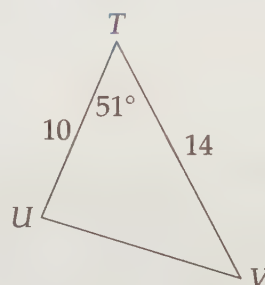
Exercises

18.3.1 Find the missing sides below to the nearest tenth.

(a)

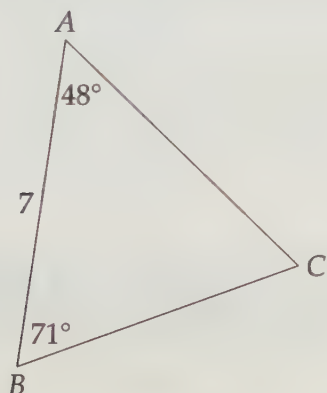


(b)

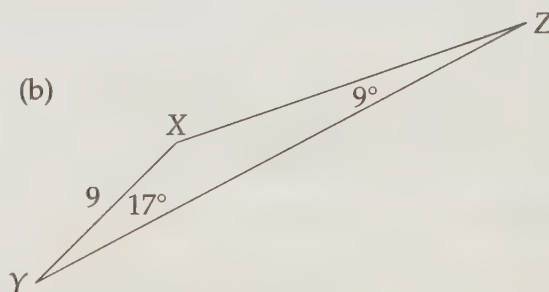


18.3.2 Find the missing sides below to the nearest tenth.

(a)



(b)



18.3.3 In this problem, we prove that the Law of Cosines works when the given angle is obtuse. Let $\triangle ABC$ be an obtuse triangle with $\angle C > 90^\circ$. Let $AB = c$, $CA = b$, and $BC = a$, as usual.

- Let D be the foot of the altitude from A to $\overleftrightarrow{BC}$. Find AD , BD , and CD in terms of a , b , c , and trigonometric functions of $\angle C$.
- Show that $c^2 = a^2 + b^2 - 2ab \cos C$.

18.3.4 Prove that the Law of Sines holds for right triangles and for obtuse triangles.

18.3.5 A surveyor is 3 kilometers from the base of a mountain. The mountain's face slopes up at an angle of 30° from horizontal. If the surveyor measures 10 degrees from horizontal to the top of the mountain, approximately how tall is the mountain to the nearest hundredth of a kilometer?

18.3.6 Let $\triangle ABC$ be a triangle with $BC = a$, $AC = b$, $AB = c$, and with circumradius R .

- Draw $\triangle ABC$ and its circumcircle. Let point D be on the circumcircle such that $\overline{AD}$ is a diameter. Find $\sin \angle ADB$ in terms of the side lengths of $\triangle ABD$.
- What angle of $\triangle ABC$ equals $\angle ADB$, and why?
- Show that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

This is the **Extended Law of Sines**.

18.3.7★ Suppose $AC = 5$, $BC = 6$, and $AB = \sqrt{31}$. Find $\angle BCA$.

18.4 Summary

We write the cosine of an angle θ as $\cos \theta$ and the sine of θ as $\sin \theta$. We define sine and cosine using the **unit circle**, which is the circle on the Cartesian plane with radius 1 centered at the origin.

Important: Let point A be $(1,0)$ and point B be on the unit circle such that B is θ degrees counterclockwise from A . Then, we define $\cos \theta$ and $\sin \theta$ to be the x -coordinate and y -coordinate, respectively, of point B .

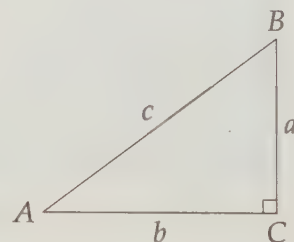
We write the tangent of an angle θ as $\tan \theta$, and define it as $\tan \theta = \frac{\sin \theta}{\cos \theta}$. Sine, cosine, and tangent are examples of **trigonometric functions**.

For an acute angle $\angle A$ of a right triangle $\triangle ABC$, we have:

$$\sin A = \frac{\text{length of leg opposite } A}{\text{length of hypotenuse}} = \frac{a}{c},$$

$$\cos A = \frac{\text{length of leg adjacent to } A}{\text{length of hypotenuse}} = \frac{b}{c},$$

$$\tan A = \frac{\text{length of leg opposite } A}{\text{length of leg adjacent to } A} = \frac{a}{b}.$$



We proved several trigonometric identities, which are statements involving trigonometric functions of angles that are true for all angles. The most important is:

Important:



$$\sin^2 A + \cos^2 A = 1$$

While we proved several other identities, it is more important to understand how these identities are derived than to memorize all the identities.

We then used trigonometry to find three important triangle relationships:

Important: Let $BC = a$, $AC = b$, and $AB = c$ in $\triangle ABC$. Then, we have



- $[ABC] = \frac{1}{2}ab \sin C$.
- $c^2 = a^2 + b^2 - 2ab \cos C$. This is called the **Law of Cosines**.
- $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. This is called the **Law of Sines**.

Problem Solving Strategies

Concepts:



- If we have the length of a side of a right triangle and the value of a trigonometric function of either of the acute angles of the triangle, then we can find all the side lengths of the triangle.
- Check formulas you prove by trying them on specific examples you have solved without the formula.
- Whenever we have to prove a general statement like the Law of Cosines, it's often useful to first consider a specific example of the general statement. If we can solve a specific example, we can sometimes use that solution as a guide to prove the generalization.

REVIEW PROBLEMS

18.24 In $\triangle ABC$ at right, find $\sin A$, $\cos A$, and $\tan A$.

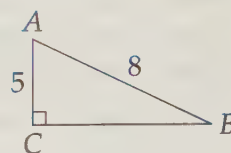
18.25 In $\triangle PQR$, let $\angle Q = 90^\circ$, $\angle P = 71^\circ$, and $PR = 16$. Find PQ and QR to the nearest tenth.

18.26 Show that if $\triangle XYZ$ is a right triangle with $\angle X = 90^\circ$, then $(\tan Y)(\tan Z) = 1$.

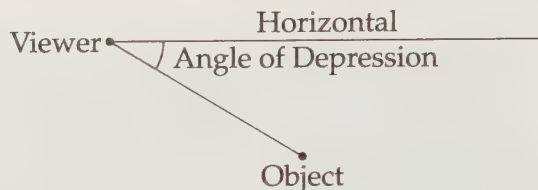
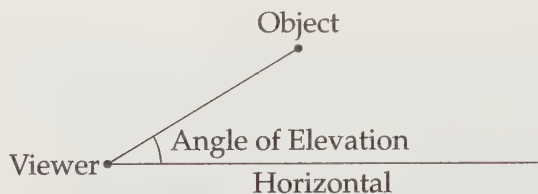
18.27 In $\triangle PQR$, we have $\angle Q = 90^\circ$ and $\sin P = 1/4$. Find $\sin R$.

18.28 Evaluate each of the following:

- | | | |
|----------------------|------------------------|----------------------|
| (a) $\sin 135^\circ$ | (c) $\cos(-120^\circ)$ | (e) $\tan 300^\circ$ |
| (b) $\sin 630^\circ$ | (d) $\cos 315^\circ$ | (f) $\tan 150^\circ$ |



18.29 The **angle of elevation** is the angle above the horizontal at which a viewer must look to see an object that is higher than the viewer. Similarly, the **angle of depression** is the angle below the horizontal at which a viewer must look to see an object that is below the viewer.



Answer each of the following problems to the nearest foot.

- (a) A surveyor measures the angle of elevation from her feet to the top of a building as 5° . The surveyor knows that the building is 500 feet tall. Assuming the ground is flat and level between the surveyor and the building, how far away is the surveyor from the building?
- (b) I'm standing at the peak of a mountain that is 14,000 feet above sea level. The angle of depression from this peak to a nearby smaller peak is 4° . On my map, these two peaks are represented by points that are 1 inch apart. If each inch on my map represents 1.2 miles, and there are 5280 feet in a mile, then how many feet above sea level is the second peak?
- (c)★ A bee is on a hill looking at a building. The building is 400 feet tall. The angle of elevation from the bee to the top of the building is 4° and the angle of depression from the bee to the bottom of the building is 2° . What is the shortest distance the bee will have to fly to reach the building?

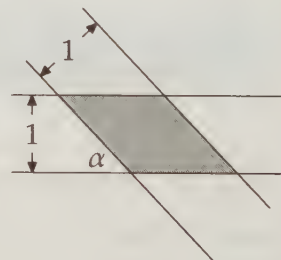
18.30 Let θ_1 and θ_2 be the two values of θ such that $0^\circ < \theta < 180^\circ$ and $\sin \theta = 0.48$. What is $\theta_1 + \theta_2$?

18.31 Suppose that $AB = 7$, $BC = 8$, and $\angle ABC = 45^\circ$. Find $[ABC]$.

18.32 If $\tan \theta$ is negative, then which quadrant(s) could θ possibly be in?

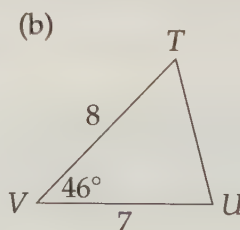
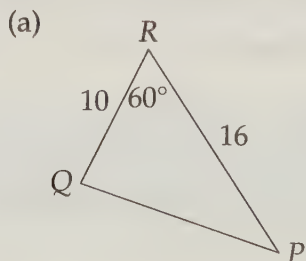
18.33 Recall that $\sec x = \frac{1}{\cos x}$. Show that for any angle x for which $\cos x \neq 0$, we have $\tan^2 x + 1 = \sec^2 x$.

18.34 Two strips of width 1 overlap at an angle of α as shown. Show that the area of the overlap (shown shaded) is $1/(\sin \alpha)$.



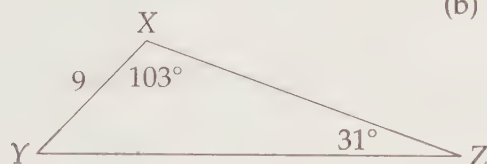
18.35 Point A is on the unit circle in the first quadrant such that A is θ degrees counterclockwise from $(1, 0)$. Point B is on the x -axis such that $\overline{AB}$ is tangent to the unit circle. Show that $\tan \theta = AB$. (Do you now see why we use the name 'tangent' for this trigonometric function?)

18.36 Find the missing side lengths in the triangles below to the nearest tenth.

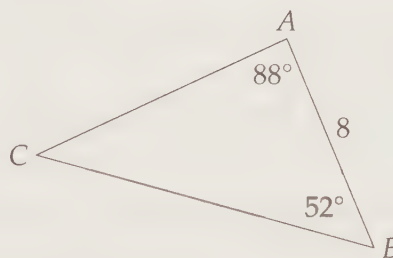


18.37 Find the missing side lengths in the triangles below to the nearest tenth.

(a)



(b)



18.38 If I walk north 3 miles, turn 32° , then walk another 2 miles, how far will I be from my initial starting point (to the nearest 0.1 mile)?

18.39 What's wrong with the following proof of the Pythagorean Theorem:

In the text, we proved that if $a = BC$, $b = AC$, and $c = AB$, then $c^2 = a^2 + b^2 - 2ab \cos C$. If $\angle C = 90^\circ$, then we have $\cos C = 0$. Therefore, if $\angle C = 90^\circ$, we have $c^2 = a^2 + b^2$.

18.40 In $\triangle PQR$, we have $PQ = 3$, $QR = 5$, and $PR = 6$.

(a) Find $\cos P$.

(b)★ Find $\sin Q$.

Challenge Problems

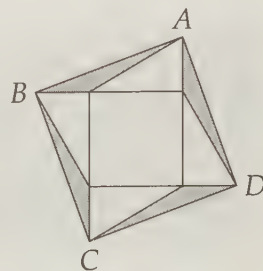
18.41 Show that if the sides of $\triangle ABC$ have lengths a , b , and c , and $\triangle ABC$ has circumradius R , then

$$[ABC] = \frac{abc}{4R}.$$

18.42 Two rays with common endpoint O form a 30° angle. Point A lies on one ray, point B on the other ray, and $AB = 1$. What is the maximum possible length of $\overline{OB}$? (Source: AHSME)

18.43 How many triangles have area 10 and vertices $(-5, 0)$, $(5, 0)$, and $(5 \cos \theta, 5 \sin \theta)$ for some angle θ ? (Source: AHSME)

18.44 Four congruent 30-60-90 triangles are constructed on the sides of a square as shown at right. The hypotenuse of each of these triangles has length 2. The outer vertices of these triangles are connected as shown to form quadrilateral $ABCD$. What fraction of $ABCD$ is shaded?



18.45 A circle centered at O has radius 1 and contains the point A . Segment $\overline{AB}$ is tangent to the circle at A and $\angle AOB = \theta$. Suppose point C is on $\overline{OA}$ such that $\overline{BC}$ bisects $\angle ABO$. Show that $OC = 1/(1 + \sin \theta)$. (Source: AMC 12)

18.46 If $\sin x = 3 \cos x$, then what is $(\sin x)(\cos x)$? (Source: AHSME)

18.47 Line ℓ intersects the x -axis at an angle of 50° . Line k makes an angle of 140° with the x -axis. The intersection of ℓ and k has a y -coordinate of 10. Find all possible values of the distance between the x -intercepts of ℓ and k .

18.48 Right triangle ABC has inradius 1 and $\sin A = 12/13$. Find the length of the hypotenuse of ABC .
Hints: 593

18.49 What's wrong with this proof of AA Similarity:

Suppose we have $\triangle ABC$ and $\triangle DEF$ with $\angle A = \angle D$ and $\angle B = \angle E$. The Law of Sines gives us

$$\frac{BC}{\sin A} = \frac{AC}{\sin B} \quad \text{and} \quad \frac{EF}{\sin D} = \frac{DF}{\sin E}.$$

Therefore, we have $BC/AC = (\sin A)/(\sin B)$ and $EF/DF = (\sin D)/(\sin E)$. Since $\angle A = \angle D$ and $\angle B = \angle E$, we have $(\sin A)/(\sin B) = (\sin D)/(\sin E)$. So, we have

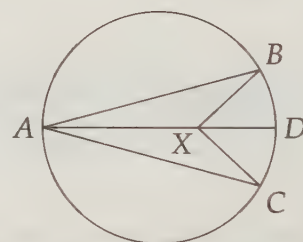
$$\frac{BC}{AC} = \frac{\sin A}{\sin B} = \frac{\sin D}{\sin E} = \frac{EF}{DF}.$$

Because $\angle A = \angle D$ and $\angle B = \angle E$, we have $\angle C = \angle F$. We can follow essentially the same steps as above to deduce $BC : AC : AB = EF : DF : DE$.

18.50★ Points A, B, C , and D are on a circle of diameter 1, and X is on diameter $\overline{AD}$. Suppose $BX = CX$ and $3\angle BAC = \angle BXC = 36^\circ$. Show that

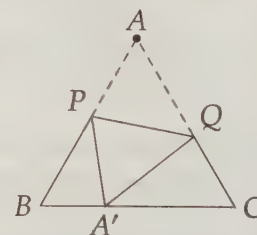
$$AX = (\cos 6^\circ)(\sin 12^\circ)/(\sin 18^\circ).$$

(Source: AHSME) Hints: 589, 576



18.51★ An object moves 8 cm in a straight line from A to B , turns at an angle of α , where $0^\circ < \alpha < 180^\circ$, and then moves 5 cm in a straight line to C . What is the probability that $AC < 7$? Hints: 526

18.52★ Equilateral triangle ABC has been creased and folded so that vertex A now rests at A' on $\overline{BC}$ as shown. If $BA' = 1$ and $A'C = 2$, then what is the length of crease $\overline{PQ}$? (Source: AMC 12) Hints: 555, 544



Extra! Since you are now studying geometry and trigonometry, I will give you a problem. A ship sails the ocean. It left Boston with a cargo of wool. It grosses 200 tons. It is bound for Le Havre. The mainmast is broken, the cabin boy is on deck, there are 12 passengers aboard, the wind is blowing East-North-East, the clock points to a quarter past three in the afternoon. It is the month of May. How old is the captain?
– Gustave Flaubert